End-Semester Exam

Numerical Algorithms

Time: 1.5 hours

Total: 100 points

2025121009

Instructions:

- Calculators are Allowed.
- Read all questions carefully.
- All questions are compulsory.

Part A: True/False (10 × 1 = 10 points)

For each statement, mark **T** (True) or **F** (False). No justification is required.

1. The forward difference $\dfrac{f(x+h)-f(x)}{h}$ approximates $f'(x)$ with truncation error of order $O(h)$. [1]

2. The central difference $\dfrac{f(x+h)-f(x-h)}{2h}$ approximates $f'(x)$ with truncation error of order $O(h^2)$. [1]

3. The composite trapezoidal rule is exact for any polynomial of degree at most 1. [1]

4. Simpson's 1/3 rule can be applied with an odd number of subintervals. [1]

5. In a convex optimization problem in standard form, equality constraints must be affine functions. [1]

6. The Lagrange dual function $g(\lambda, \nu)$ is concave in (λ, ν), even if the primal problem is not convex. [1]

7. Weak duality (for a minimization primal) states that the optimal dual value is always greater than or equal to the optimal primal value. [1]

8. If strong duality holds and (x^*, λ^*, ν^*) is optimal, then complementary slackness gives $\lambda_i^* f_i(x^*) = 0$ for every inequality constraint. [1]

9. Experience replay is required for tabular Q-learning to converge. [1]

10. The Bellman optimality equation for Q^* includes a $\max_{a'} Q^*(s', a')$ term, which motivates the Q-learning update. [1]

Part B: Short Answer ($6 \times 5 = 30$ points)

Answer briefly and clearly. Show key steps where requested.

11. **Standard form and feasible set.** Consider the problem (variable $x \in \mathbb{R}$):

$$\begin{aligned} \text{minimize} \quad & (x-1)^2 + 2 \\ \text{subject to} \quad & 2x - 3 \geq 0, \\ & x \leq 4. \end{aligned}$$

   (a) Rewrite it in standard form: minimize $f_0(x)$ subject to $f_i(x) \leq 0$. [3]

   (b) Write the feasible set explicitly as an interval. [2]

12. **Convexity check.** Let $f(x) = -\log x$ with domain $(0, \infty)$.

   (a) Compute $f''(x)$ and use it to show f is convex on $(0, \infty)$. [4]

   ? (b) State the domain of f clearly. [1]

13. **Conjugate and reflexivity (biconjugate).**

   (a) Define the Fenchel conjugate $f^*(y)$ of a function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$. [2]

   (b) Compute the conjugate of $f(x) = \frac{1}{2}x^2$ (one-dimensional). [2]

   (c) Compute $f^{**}(x)$ for this f and state whether $f^{**} = f$. [1]

14. **Weak duality and complementary slackness.**

   (a) In 3–6 lines, prove weak duality using the Lagrangian lower-bound idea (for a minimization primal). [3]

   (b) State complementary slackness for inequality constraints. [2]

15. **RL fundamentals and Bellman equations.**

   (a) Define the state-value function $V^\pi(s)$ and action-value function $Q^\pi(s, a)$. [2]

   (b) Write the Bellman expectation equation for $V^\pi(s)$ in an MDP (discount $\gamma \in [0, 1)$). [2]

   ? (c) In 1–2 sentences: why does Q-learning use a Bellman equation? [1]

? 16. **State/action representations in RL.**

   ? (a) Give a reasonable *state representation* and *action set* for **chess**. [2]

   ? (b) Give a reasonable *state representation* and *action set* for **cart-pole**. [1]

   ? (c) Give a reasonable *state representation* and *action set* for **tetris/brick game**. [1]

   ? (d) For **Atari from pixels**, how are states commonly represented, and what is the role of a CNN? [1]

Part C: Long Answer ($3 \times 20 = 60$ points)

Show all essential steps. Numerical answers should be clearly labeled.

17. Numerical differentiation, numerical integration, and Newton's method. Let $f(x) = e^x$ and use step size $h = 0.1$.

(a) Approximate $f'(0.2)$ using:

$$\text{(i) forward: } \frac{f(0.2+h) - f(0.2)}{h}, \qquad \text{(ii) central: } \frac{f(0.2+h) - f(0.2-h)}{2h}.$$

Report both approximations (at least 4 decimal places). [6]

(b) Approximate $f''(0.2)$ using the second-order central difference:

$$f''(0.2) \approx \frac{f(0.2+h) - 2f(0.2) + f(0.2-h)}{h^2}.$$

Compute the exact value $f''(0.2)$ and the absolute error. [4]

(c) Use **composite Simpson's rule** with $n = 4$ subintervals (so $h = 0.1$) to approximate

$$I = \int_0^{0.4} e^x \, dx.$$

Compute the exact value $I_{\text{exact}} = e^{0.4} - 1$ and the absolute error. [5]

(d) Consider minimizing $g(x) = e^x - 2x$ in one dimension. Newton's method for minimization is

$$x_{k+1} = x_k - \frac{g'(x_k)}{g''(x_k)}.$$

Starting from $x_0 = 0$, compute x_1, x_2, x_3. What value should the iterates converge to? [5]

18. Convex optimization, duality, and KKT. Consider the problem with variable $x = (x_1, x_2) \in \mathbb{R}^2$:

$$\begin{aligned} \text{minimize} \quad & \frac{1}{2}(x_1^2 + x_2^2) \\ \text{subject to} \quad & x_1 + x_2 \geq 1, \\ & x_1 \geq 0, \quad x_2 \geq 0. \end{aligned}$$

(a) Rewrite the constraints in **standard form** $f_i(x) \leq 0$ and clearly list f_0, f_1, f_2, f_3. [4]

? (b) Is this a standard **convex optimization problem**? Justify briefly (objective/constraints). Also state Slater's condition and whether it holds here. [4]

(c) Form the Lagrangian $L(x, \lambda)$ (use multipliers $\lambda_1, \lambda_2, \lambda_3 \geq 0$ for the three inequalities). Minimize L over x to derive an explicit dual function $g(\lambda)$. [6]

? (d) Write the **dual problem** and solve it (find λ^* and the dual optimal value d^*). [4]

? (c) Recover a primal optimal solution x^* from λ^* and verify complementary slackness for all constraints. [2]

19. Q-learning on a grid world (no experience replay). Consider a deterministic 2×2 grid:

$$s_1 = \text{top-left (start)}, \quad s_2 = \text{top-right}, \quad s_3 = \text{bottom-left}, \quad s_T = \text{bottom-right (terminal)}.$$

Actions are $\{R, D\}$ (Right, Down). If an action would leave the grid, the state does not change. Rewards are defined by:

- If the transition enters s_T, reward $= +5$ and the episode ends.
- Otherwise reward $= -1$.

Use discount $\gamma = 0.9$, learning rate $\alpha = 0.5$, and initialize $Q_0(s, a) = 0$ for all (s, a).

? (a) Write the tabular Q-learning update rule and explain the meaning of each term in one short sentence. [4]

(b) The agent experiences the following two episodes (same behavior both times):

$$\text{Episode 1: } s_1 \xrightarrow{R,-1} s_2 \xrightarrow{D,+5} s_T, \qquad \text{Episode 2: } s_1 \xrightarrow{R,-1} s_2 \xrightarrow{D,+5} s_T.$$

Perform Q-learning updates **in chronological order** (update immediately after each transition). Compute the value of $Q(s_1, R)$ and $Q(s_2, D)$ **after each update**. [10]

(c) After Episode 2, give the greedy action (argmax over actions) at s_1 and at s_2. Also compute $V(s_1) = \max_a Q(s_1, a)$ and $V(s_2) = \max_a Q(s_2, a)$. [2]

? (d) What is experience replay? Give one reason it can improve stability/sample-efficiency in deep Q-learning with function approximation. [2]

? (e) State the Bellman optimality equation for Q^*. Then give two standard conditions under which tabular Q-learning converges to Q^* (e.g., learning-rate and visitation/exploration conditions), and briefly mention why the Bellman optimality operator matters for convergence. [2]

End of Exam